

Sample Project Documentation

Smart Lender - Loan Eligibility Prediction System

Project Description

Smart Lender is a machine learning-based web application that predicts whether an applicant is eligible for a loan based on details such as income, education, employment, loan amount, credit history, and other financial parameters. The system helps banks and financial institutions make faster, consistent, and data-driven loan approval decisions.

Key Features

- Loan eligibility prediction using Machine Learning
- User-friendly web interface
- Applicant data validation
- Real-time prediction results
- Model trained on historical loan data
- Easy deployment using Streamlit

Technology Stack

Programming Language: Python

Framework: Streamlit

Libraries: Scikit-learn, Pandas, NumPy

Model: Random Forest / Logistic Regression

Project Workflow

Milestone 1: Collect and preprocess loan dataset

Milestone 2: Train and evaluate ML model

Milestone 3: Develop Streamlit interface

Milestone 4: Integrate prediction model

Milestone 5: Test application

Milestone 6: Deploy application